Joux–Vitse $(n-1)$-decomposition at $n=5$: measured against rho

2026-10-15 · `research/joux_vitse_n1_20261015/` · code `src/cryptanalysis/joux_vitse.rs`, `examples/joux_vitse_ladder.rs`, summary by `examples/joux_vitse_summary.rs` · protocol `PROTOCOL.md` (frozen before data) and `PROTOCOL_ADDENDUM.md` · cost model `MODEL.md`. **Class: measurement of a published route (reproduction), with an accounting correction to the scoreboard's $k=5$ plain constant; no advance.**

Verdict

No. On $E(\mathbb{F}_{q^5})$ the $(n-1)$-decomposition costs about $12q^2$ trials of about $7.53{\cdot}10^9$ $\mathbb{F}_p$ multiplications each (measured, flat in q), against rho's $\approx 0.89q^{5/2}$ group operations of 190 multiplications. Measured end to end on complete, verified DLPs it is $4.5\cdot10^7$–$1.8\cdot10^8$ times rho at $q=11,31$; on the measured constants the extrapolated crossover is $q^*\approx 2^{58.2}$ (group $\approx 2^{291}$), consistent with Vitse's "faster from a base field larger than 2^{60}" (thesis §7.3.3). On the model, IC is still $\approx 10^7$ times rho at $q=2^{10}$ and $\approx 10^6$ at $q=2^{16}$ (extrapolation).

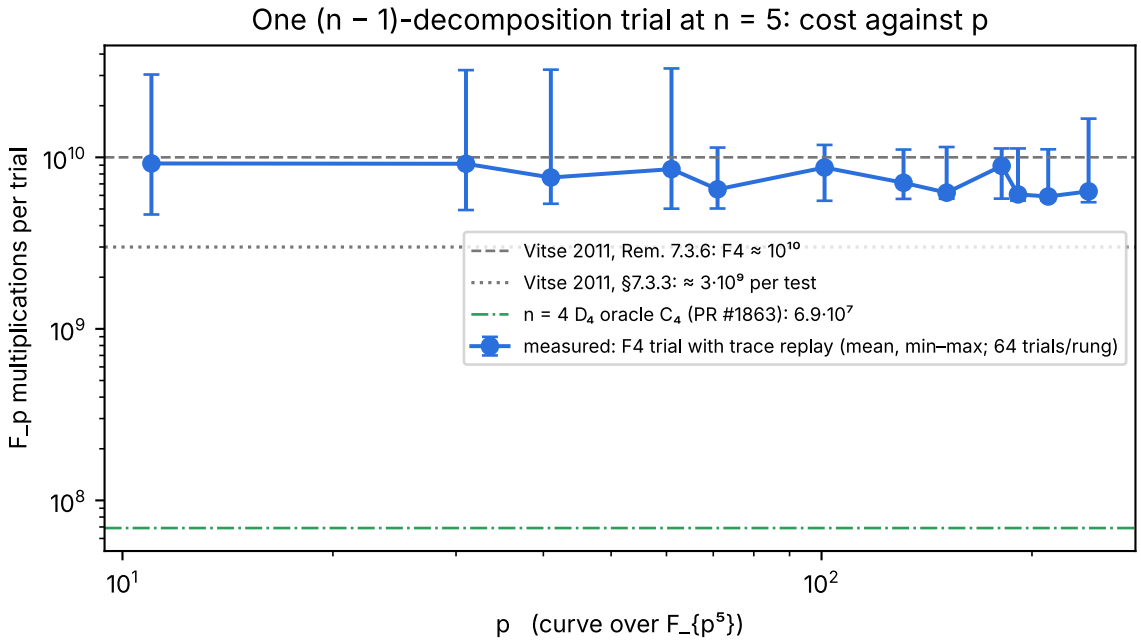
Literature, and two corrections to the premise

The practical $\mathbb{F}_{p^6}$ result of Joux–Vitse (Eurocrypt 2012) is *cover and decomposition* (a genus-2/3 cover over an intermediate field, Nagao decompositions), not $(n-1)$-decomposition on $E(\mathbb{F}_{p^6})$; the practical $\mathbb{F}_{q^5}$ result of the $n-1$ variant is an *oracle-assisted static Diffie–Hellman* attack (Oakley curve, $\mathbb{F}_{(2^{31})^5}$), not a DLP. The thesis prices one trial at $\approx 3\cdot10^9$ multiplications (F4: 10^{10}; F4Remake about 3× fewer) and the crossover with rho at $q\approx 2^{60}$ (odd p). The repository already had the plain and 2-torsion $k=5$ test as a stage diagnostic (`jv_quintic.rs`, RESEARCH_K5_TORSION_JOUX_VITSE.md); this study duplicates the plain construction (an incomplete survey, recorded) and adds the trace replay and the first complete $k=5$ DLPs.

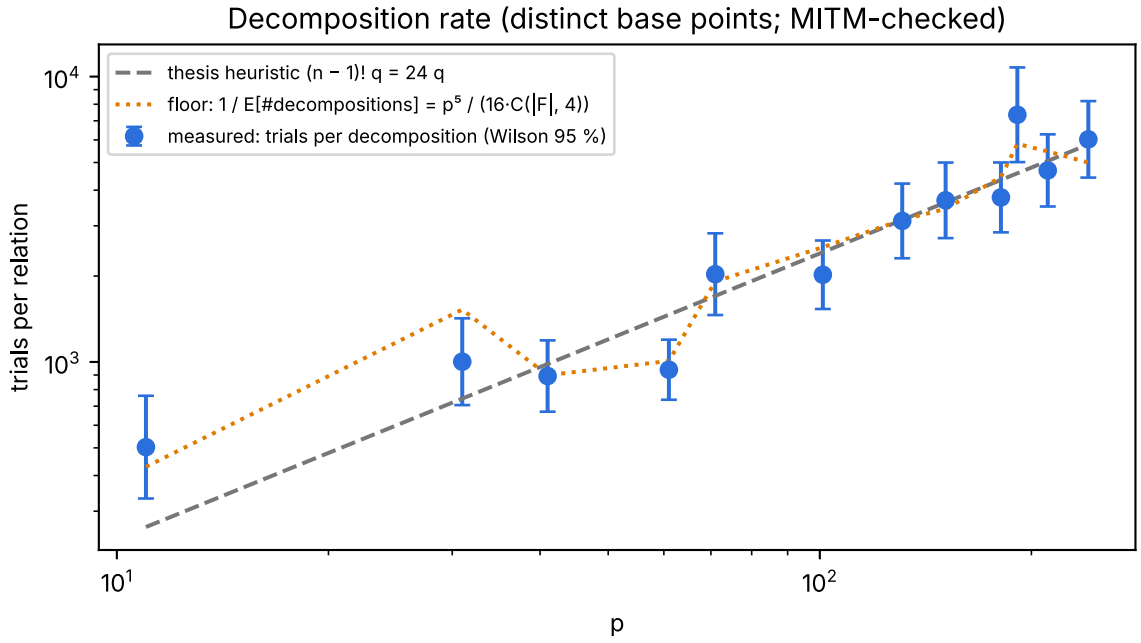
One trial, across p (12 rungs, 2 curves × 32 F4 trials each)

p	$\lvert F\rvert$	C_{trial} mean	min	max	replay div.	hits / trials	trials/rel.	95 %	rate·24p	C_{rel}
11	6.5	$9.21{\cdot}10^9$	$4.64{\cdot}10^9$	$3.04{\cdot}10^{10}$	5	22 / 11064	503	332–761	0.52	$4.63{\cdot}10^{12}$
31	14.5	$9.17{\cdot}10^9$	$4.93{\cdot}10^9$	$3.22{\cdot}10^{10}$	3	31 / 31064	1002	706–1422	0.74	$9.19{\cdot}10^{12}$
41	22.5	$7.63{\cdot}10^9$	$5.36{\cdot}10^9$	$3.24{\cdot}10^{10}$	2	46 / 41064	893	669–1190	1.1	$6.81{\cdot}10^{12}$
61	35	$8.54{\cdot}10^9$	$5.02{\cdot}10^9$	$3.3{\cdot}10^{10}$	4	65 / 61064	939	737–1197	1.56	$8.02{\cdot}10^{12}$
71	36	$6.51{\cdot}10^9$	$5.03{\cdot}10^9$	$1.14{\cdot}10^{10}$	0	35 / 71064	2030	1460–2823	0.84	$1.32{\cdot}10^{13}$
101	51.5	$8.72{\cdot}10^9$	$5.58{\cdot}10^9$	$1.18{\cdot}10^{10}$	0	50 / 101064	2021	1533–2664	1.2	$1.76{\cdot}10^{13}$
131	67	$7.11{\cdot}10^9$	$5.72{\cdot}10^9$	$1.11{\cdot}10^{10}$	0	42 / 131064	3121	2309–4218	1.01	$2.22{\cdot}10^{13}$
151	78	$6.22{\cdot}10^9$	$5.74{\cdot}10^9$	$1.15{\cdot}10^{10}$	0	41 / 151064	3684	2716–4998	0.98	$2.29{\cdot}10^{13}$
181	91.5	$8.91{\cdot}10^9$	$5.74{\cdot}10^9$	$1.13{\cdot}10^{10}$	0	48 / 181064	3772	2845–5001	1.15	$3.36{\cdot}10^{13}$
191	91.5	$6.08{\cdot}10^9$	$5.59{\cdot}10^9$	$1.13{\cdot}10^{10}$	0	26 / 191064	7349	5015–10768	0.62	$4.47{\cdot}10^{13}$
211	105	$5.91{\cdot}10^9$	$5.61{\cdot}10^9$	$1.11{\cdot}10^{10}$	0	45 / 211064	4690	3506–6275	1.08	$2.77{\cdot}10^{13}$
241	126.5	$6.33{\cdot}10^9$	$5.47{\cdot}10^9$	$1.68{\cdot}10^{10}$	0	40 / 241064	6027	4426–8206	0.96	$3.82{\cdot}10^{13}$

Fits (least squares on logs): $C_{\text{trial}}\propto p^{-0.12}$ (± 0.04); trials per relation $\propto p^{0.87}$ (± 0.09); $C_{\text{rel}}\propto p^{0.75}$ (± 0.08). Checks failed: **0** (every F4 trial's decompositions equal the meet-in-the-middle set; every relation re-adds to R). Replayed trials cost one of three levels (≈ 5.7, 8.2, $11.1\cdot10^9$); the suspected cause is `f4_fp` finishing pending pairs as a full run after the recorded steps (not verified per trial). The committed `jv_quintic` plain test re-run on today's engine costs $1.104\cdot10^{10}$ (it was $3.0\cdot10^{10}$ on 2026-09-29): the engine's own progress, not this study's.



Cost of one trial ($\mathbb{F}_p$ multiplications) against p; references from Vitse's thesis and PR #1863's $n = 4$ D_4 oracle (different group).

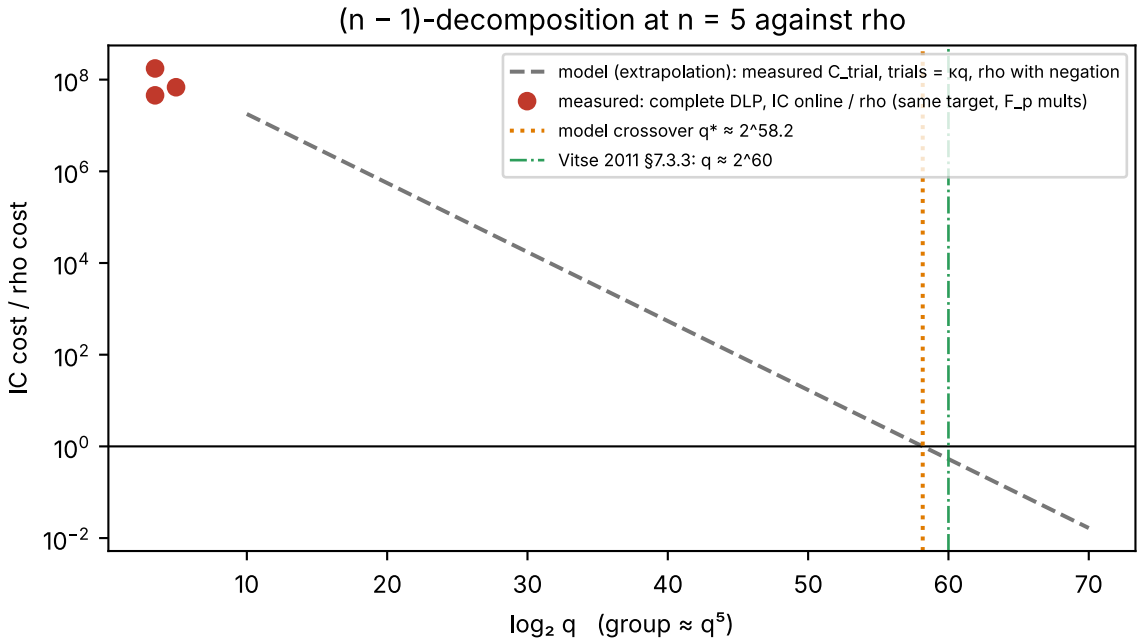


Trials per decomposition (Wilson 95 %), the thesis heuristic $24q$, and the counting floor $p^5/\left(16\binom{|F|}{4}\right)$ computed from each rung's base.

Complete DLPs, paired with rho on the same target

run	ℓ	IC trials	relations	IC online ($\mathbb{F}_p$ mult.)	rho ops	rho $\mathbb{F}_p$ mult.	IC / rho	IC / exp. walk	verified
p11_s1.jsonl	160637	1912	6	$1.74{\cdot}10^{13}$	2030	$3.84{\cdot}10^{5}$	$4.54{\cdot}10^{7}$	$1.84{\cdot}10^{8}$	both
p11_s2.jsonl	4493	2920	7	$2.67{\cdot}10^{13}$	800	$1.52{\cdot}10^{5}$	$1.75{\cdot}10^{8}$	$1.67{\cdot}10^{9}$	both
p11_s3.jsonl	14669	stopped	0	—	1054	$2{\cdot}10^{5}$	—	—	rho only
p31_s1.jsonl	28622519	6688	19	$5.31{\cdot}10^{13}$	4092	$7.78{\cdot}10^{5}$	$6.82{\cdot}10^{7}$	$4.16{\cdot}10^{7}$	both

Rho's counts include its 20 multipliers' set-up, which dominates at the $p = 11$ orders; the last ratio column prices rho as its expected walk alone, $\sqrt{\pi\ell/2}$ operations at the measured cost of one. At $p = 31$ the measured walk was short (4,092 operations against 6,705 expected), so there the expected-walk ratio is the smaller one. $p = 11$, seed 3 drew a curve with $|F| = 1$ and was stopped (`data/dlp/p11_s3.STOPPED.md`). Target-independent set-up (symmetrised S_5, $1.87 \cdot 10^9$; trace recording, $\approx 10^{10}$) is reported separately and excluded from "online".



IC / rho: the complete DLPs (red) and the model (dashed; extrapolation on the measured C_{trial}, trials $= \kappa q$ with $\kappa = 25.4$, rho with the negation map).

$n = 6$, and what was not done

$n = 6$ was not attempted: the symmetrised S_6 has 20,349 monomials × 17 powers of x_R (a dense system of ≈ 20 GB against this host's 15 GB, $\approx 1.3 \cdot 10^{14}$ multiplications), and one trial is six degree-16 equations in five unknowns, Macaulay bound 91 (MODEL.md; derived, a resource limit). No ecbench run: `main` has no $\mathbb{F}_{p^k}$ instance (`Instance::Fp4` is in the unmerged #1792), recorded in `problems/k5-joux-vitse-trial-cost-and-fpk-ecbench.md`. The curves' ICV1 identities are computed (`curves.json`) but not registered (the registry builder harvests no extension-field JSON).

Environment: Claude Code cloud container, Intel Xeon @ 2.10 GHz, 4 vCPU, 15 GB, rustc 1.97.0, commit `1e8592da0` for every frozen run; isolation none (wall times are practicality notes only). A container restart killed the first launch; its partial records are kept in `data_killed_20261015T0518Z/` and the restarted runs reproduced their relations exactly.